

Fig. 13: Options for Target window

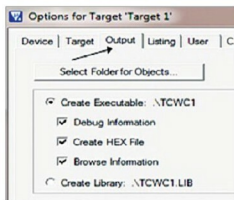


Fig. 14: Create hex file window

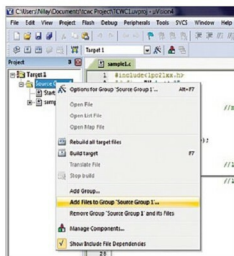


Fig. 15: Add Files window

For troubleshooting, check voltages at various points mentioned in the test points table.

Software

The software is written in embedded C language and compiled using Keil Micro vision 4 Version Compiler. It

contains simple branch statements to control different fan speeds. We have used TopWin 850 programmer to burn the hex code into the MCU. Follow the following steps to compile the program.

Step 1. After opening Keil uVision, go to Project

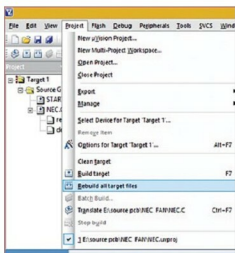


Fig. 16: Build Target Options window

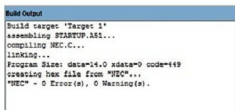


Fig. 17: Output window

and create new uVision project. Select New Folder and give a name to the Project as shown in Fig. 6.

Step 2. After creating a project, select your device model as shown in Fig. 7. For example, AT89C2051. It is available in Atmel package.

Step 3. Now your project is created; a message window will appear to add startup file of your device. Click on Yes to add to your project folder as shown in Fig. 8.

Step 4. Go to File and create a

EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download at source.efymag.com

Test Points

Test point	Details
TP0	0V
TP1	12V
TP2	5V

new file. Save it with .c extension if you have written the program in C language, or save with .asm for Assembly language as shown in Fig. 9.

Step 5. Write your program and save it again, as shown in Fig. 10.

Step 6. On the left you will see the project window. If not, go to View and click on Project Window as shown in Fig. 11.

Step 7. After clicking Project Window, right-click on Target and click on Options for Target (Fig. 12).

Step 8. Click Output tab and check Create Hex File. Click on OK, so it saves all changes.

Step 9. Expand Target. You will see Source Group. Right-click on Group and click on Add Files to Source Group.

Here, add your program file that you have written in C language. The program file is added under the source group.

Step 10. Click on Build Target. You will find it under Project or in the toolbar. This can also be done by pressing F7 key.

Step 11. You can see the status of your program in Build Output window. If so, you are done with your program. **EFY**



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